

METHODS AND CONSTRUCTIONS FOR RAMSEY THEORY

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ABSTRACT. Ramsey Theory is an active area of research in mathematics which formulates extremal coloring properties of graphs. This paper studies different combinatorial methods, both probabilistic and geometric, to construct particular families of graphs and obtain bounds on their corresponding Ramsey numbers.

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1. INTRODUCTION

Throughout the paper $n \in \mathbb{N}$, $q > 2$ is a prime power, $\mathbb{F}_q$ denotes the unique finite field of order q , $[n] := \{1, 2, \dots, n\}$, $\mathcal{S}_n$ is the symmetric group of order n and for a finite set A we denote $\#A$ the cardinality of A . Other notations will be introduced in later sections.

1.1. History and basic definitions. Ramsey Numbers were first introduced by Frank.P Ramsey in the early 20th century and was the beginning of an entire area of combinatorics called Ramsey Theory. Let $n, k_1, \dots, k_n \in \mathbb{N}$, we define the generalized Ramsey number $R(k_1, \dots, k_n)$ to be the smallest natural number such that given any edge coloring¹ of the complete graph $K_{R(k_1, \dots, k_n)}$ in n colors, there exists $i \in [n]$ and a complete monochromatic subgraph of size k_i . Ramsey showed that these numbers always exist, i.e $R(k_1, \dots, k_n) < \infty$, which is done quite simply by a pigeon hole argument. However, computing Ramsey Numbers is an extremely hard task and so far no one has been able to compute these numbers reliably. The best we have done for the simplest case $n = 2$ is computing up to $R(4, 4)$ as is shown in 1. Studying the asymptotic and quantitative behavior of $R(s, t)$ has been a question leading to extensive research, initiated by Paul Erdős in his 1947 paper [1]. Since most of the best lower bounds have come from probabilistic arguments, explicit examples of colorings without any complete monochromatic subgraphs of size s and t are rare. Producing such examples can be very insightful in illustrating how counter examples arise and their typical structure.

TABLE 1. Values and bounds for $R(r, s)$

$r \backslash s$	1	2	3	4	5	6	7	8	9	10
1	1	1	1	1	1	1	1	1	1	1
2		2	3	4	5	6	7	8	9	10
3			6	9	14	18	23	28	36	40–41
4				18	25	36–40	49–58	59–79	73–105	92–135
5					43–46	59–85	80–133	101–193	133–282	149–381
6						102–160	115–270	134–423	183–651	204–944
7							205–492	219–832	252–1368	292–2119
8								282–1518	329–2662	343–4402
9									565–4956	581–8675
10										798–16064

1.2. About the paper. The paper first introduces required tools which will come in helpful for the construction of relevant graphs. We begin by describing the probabilistic method, where we primarily explore insightful

¹An edge coloring in n colors of a graph $G = (V, E)$ is defined formally as a function $\gamma : E \rightarrow [n]$ and a graph is monochromatic if for some $i \in [n]$ we have $\gamma(E) = \{i\}$.

examples suitable for later chapters. We examine the spectrum of graphs which will be crucial in the application of geometry to Ramsey theory. A family of graphs, known as strongly regular graphs, will be of particular interest as they will naturally appear in finite geometry. Then we study the so-called incidence geometries or finite geometries which have an incredibly regular structure and give rise to graphs with desired properties. Finally, we bring all the tools together to provide a detailed study of bounds on Ramsey numbers.

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2. THE PROBABILISTIC METHOD

The Probabilistic Method is a proof technique, pioneered in the 20th century by Paul Erdős, which is used in a variety of mathematical theories. The method is used to prove the existence of objects with certain properties, which is stated as follows:

Let $(\Omega, \mathbb{P})$ be a finite probability space and $A \subset \Omega$. If $\mathbb{P}[A] > 0$, then $A \neq \emptyset$, or equivalently $\exists a \in A$.

Intuitively, this means if an object has a non-zero probability of occurring then it must exist. The method thus provides a non-constructive way of producing objects with desired properties. Furthermore, transforming a problem into a random setting allows us to use many powerful tools from probability theory, such as the expectation of random variables, Markov's inequality and Chebyshev's inequality, which will turn to be extremely useful. To introduce the method, which we will later use to give bounds on Ramsey numbers, let us look at a variety of examples from different fields of mathematics.

2.1. Sets of lines in $\mathbb{F}_q^3$. The following example provides a bound on the size of a set of lines which satisfies a stability property.

Theorem 2.1. *Let $E = \mathbb{F}_q^3$ and S a set of 1-dimensional subspaces of E such that for any $g \in \text{GL}(E)$ we have that $g(S) \cap S \neq \emptyset$, then $\#S \geq q + 1$.*

Proof. We suppose that $\#S \leq q$ and consider $g \in \text{GL}(E)$ uniformly at random, then:

$$\mathbb{P}[g(S) \cap S \neq \emptyset] = \mathbb{P}\left[\bigcup_{L \in S} \{L \in g(S)\}\right] \leq \sum_{L \in S} \mathbb{P}[L \in g(S)] = \sum_{L \in S} \#S \cdot \mathbb{P}[L = g(s)]$$

where s is taken in S . Noting that the number of lines in E is $\frac{q^3-1}{q-1}$ and that $g(s)$ is uniformly distributed on the set of 1 dimensional spaces, we conclude

that $\mathbb{P}[L = g(s)] = \frac{q-1}{q^3}$. We are now ready to finish the proof:

$$\mathbb{P}[g(S) \cap S \neq \emptyset] \leq \sum_{L \in S} \#S \cdot \mathbb{P}[L = g(s)] = \sum_{L \in S} \#S \cdot \frac{q-1}{q^3} \leq \frac{q^2(q-1)}{q^3} < 1$$

Thus, there must exist some isomorphism, g , such that $g(S) \cap S = \emptyset$. Taking the contrapositive form gives the result. $\square$

Remark 2.2. The above bound is tight. In a general setting, for any plane P of a 3-dimensional vector space E , denote S the set of 1-dimensional subspaces of P . For any $g \in \text{GL}(E)$, since $g(P)$ is still a plane of E , and intersects P on a line it follows $g(S) \cap S \neq \emptyset$. Now taking the plane $P := \{(x, y, 0) \in \mathbb{F}_q^3\}$, the number of 1-dimensional subspaces contained in P is exactly $q+1$, providing a case of equality.

Remark 2.3. We could have equivalently formulated the theorem over the classical projective plane. If we consider the classical projective plane over $\mathbb{F}_q$ and a set of points $S \subset \mathcal{P}$ such that for any $g \in \text{PGL}(\mathbb{F}_q^3)$ we have $g(S) \cap S \neq \emptyset$, then the size of S is at least $p+1$. More generally we could ask the question for any finite geometry $G = (\mathcal{P}, \mathcal{L}, \mathcal{I})$ where we take $g \in \text{Aut}(G)$, for example over Affine Planes or Generalized Quadrangles.

2.2. The Expectation. Before the next examples, we provide a tool using the expectation. If X is some real random variable over a finite probability space $(\Omega, \mathbb{P})$ and $\mu := \mathbb{E}[X]$ then $\exists \omega \in \Omega$ such that $X(\omega) \geq \mu$ and similarly $\exists \nu \in \Omega$ such that $X(\nu) \leq \mu$. Clearly if the first statement weren't the case then $\mathbb{E}[X] = \sum_{\omega \in \Omega} X(\omega) \mathbb{P}(\omega) < \mu \sum_{\omega \in \Omega} \mathbb{P}(\omega) = \mu$ which is a contradiction and a symmetric argument proves the second statement. We also remind the reader that the expectation is a linear operator and if $X : \Omega \rightarrow \{0, 1\}$ is a random indicator variable then the $\mathbb{E}[X] = \mathbb{P}[X = 1]$.

2.3. Hamiltonian Paths in Tournaments. We provide an example due to Szele [2], which is often considered the first ever use of the probabilistic method.

Definition 2.4. A tournament on n vertices, is a tuple $T = (\{v_1, \dots, v_n\}, E)$ where $E \subset \{v_1, \dots, v_n\}^2$ is such that $\forall i \in [n]$, $(v_i, v_i) \notin E$ and for each pair $(i, j) \in [n]^2$, $i \neq j$ one of $(v_i, v_j), (v_j, v_i)$ is in E but not both. In other words it is an orientation of the edges in K_n .

Theorem 2.5. *There exists a tournament on n vertices with at least $n! \cdot 2^{-(n-1)}$ Hamiltonian paths.*

Proof. We consider a random tournament on n vertices, $T = (\{v_1, \dots, v_n\}, E)$, where for each pair $i \neq j$, the events $(v_i, v_j) \in E$ and $(v_j, v_i) \in E$ are equally likely. Denote X the number of Hamiltonian paths in T and for a permutation σ on $\{v_1, \dots, v_n\}$ we write $X_\sigma := \mathbb{1}[(v_{\sigma(1)}, \dots, v_{\sigma(n)}) \text{ is Hamiltonian}]$.

Observing that $X = \sum_{\sigma \in \mathcal{S}_n} X_\sigma$ and applying the linearity of the expectation:

$$\mathbb{E}[X] = \sum_{\sigma \in \mathcal{S}_n} \mathbb{E}[X_\sigma] = \sum_{\sigma \in \mathcal{S}_n} \mathbb{P}[(v_{\sigma(1)}, \dots, v_{\sigma(n)}) \text{ is Hamiltonian}] = \sum_{\sigma \in \mathcal{S}_n} 2^{-(n-1)}$$

Thus, there must exist a tournament T with at least $n! \cdot 2^{-(n-1)}$ Hamiltonian paths. $\square$

2.4. Hadamard's determinant problem. A famous problem posed by Jacques Hadamard is the following:

How large can $|\det(A)|$ be for real $n \times n$ matrices with coefficients in $\{-1, 1\}$?

Theorem 2.6. *There exists a real matrix A with entries in $\{-1, 1\}$ such that $|\det(A)| \geq \sqrt{n!}$.*

Proof. Let X be a $n \times n$ matrix with entries in $\{-1, 1\}$ chosen uniformly at random. We shall compute $\mathbb{E}[\det(X)^2]^2$:

$$\begin{aligned} \mathbb{E}[\det(X)^2] &= \frac{1}{2^{n^2}} \sum_{A \in M_n(\{-1, 1\})} \det(A)^2 \\ &= \frac{1}{2^{n^2}} \sum_A \sum_{\sigma, \tau \in \mathcal{S}_n} \varepsilon(\sigma) \varepsilon(\tau) \prod_{i=1}^n a_{i, \sigma(i)} a_{i, \tau(i)} \\ &= \frac{1}{2^{n^2}} \sum_{\sigma, \tau \in \mathcal{S}_n} \varepsilon(\sigma) \varepsilon(\tau) \sum_A \prod_{i=1}^n a_{i, \sigma(i)} a_{i, \tau(i)} \end{aligned}$$

Now noting that the sum over $A \in M_n(\{-1, 1\})$ is equivalent to summing over all n^2 variables we find:

$$\mathbb{E}[\det(X)^2] = \frac{1}{2^{n^2}} \sum_{\sigma, \tau \in \mathcal{S}_n} \varepsilon(\sigma) \varepsilon(\tau) \sum_{a_{1,1}=\pm 1} \sum_{a_{1,2}=\pm 1} \cdots \sum_{a_{n,n}=\pm 1} \prod_{1 \leq i \leq n} a_{i, \sigma(i)} a_{i, \tau(i)}$$

Fix σ, τ and suppose $\exists j \in \{1, \dots, n\}$ such that $k := \sigma(i) \neq \tau(i)$. Then the inner sum must be zero since it can be split into the two cases where $a_{i,k} = 1$ and $a_{i,k} = -1$ which both give the same result with just a difference of sign.

²This value is called the squared mean of $\det(X)$. The motivation to consider this value is simply because $\mathbb{E}[X] = \sum_{A \in M_n(\{-1, 1\})} \sum_{\sigma \in \mathcal{S}_n} \varepsilon(\sigma) \prod_{i=1}^n \mathbb{E}[a_{i, \sigma(i)}] = 0$.

We get the following simplification:

$$\begin{aligned}
\mathbb{E}[\det(X)^2] &= \frac{1}{2^{n^2}} \sum_{\sigma, \tau \in \mathcal{S}_n} \varepsilon(\sigma)^2 \sum_{a_{1,1}=\pm 1} \sum_{a_{1,2}=\pm 1} \cdots \sum_{a_{n,n}=\pm 1} \prod_{i=1}^n a_{i,\sigma(i)}^2 \\
&= \frac{1}{2^{n^2}} \sum_{\sigma, \tau \in \mathcal{S}_n} \sum_{a_{1,1}=\pm 1} \sum_{a_{1,2}=\pm 1} \cdots \sum_{a_{n,n}=\pm 1} 1 \\
&= \frac{1}{2^{n^2}} \sum_{\sigma \in \mathcal{S}_n} 2^{n^2} = n!
\end{aligned}$$

Allowing us to conclude by the probabilistic method that $\exists A \in M_n(\{-1, 1\})$ such that $\det(A)^2 \geq n!$ and thus $|\det(A)| \geq \sqrt{n!}$. $\square$

A tight upper bound for the Hadamard problem is also known. Let $A \in M_n(\{-1, 1\})$ and we consider the real symmetric matrix tAA . By the Spectral Theorem the matrix is diagonalizable and has real eigenvalues $\lambda_1, \dots, \lambda_n$, hence, $\det({}^tAA) = \prod_{i=1}^n \lambda_i$. By AM-GM we find $\det({}^tAA) = \prod_{i=1}^n \lambda_i \leq \left(\frac{\text{tr}({}^tAA)}{n}\right)^n$ and observing $\text{tr}({}^tAA) = \sum_{i,j=1}^n a_{i,j}^2 = n^2$ we may conclude $|\det(A)| \leq n^{\frac{n}{2}}$. Matrices which attain this upper bound are called Hadamard matrices. Examples of Hadamard matrices exists for all powers of two. Let $m := 2^n$ and consider $\{S_1, S_2, \dots, S_m\}$ the power set of $[n]$ then $A := (a_{i,j})_{1 \leq i,j \leq m}$ with $a_{i,j} := (-1)^{\#S_i \cap S_j}$ is a Hadamard matrix.

From the Stirling approximation formula we have $\sqrt{n!} \sim n^{\frac{n}{2}} \frac{(2\pi n)^{\frac{1}{4}}}{e^{\frac{n}{2}}}$ which is relatively close to the determinant of Hadamard matrices, the correct growth as $n \rightarrow \infty$ is still unknown.

3. SPECTRAL GRAPH THEORY

Spectral Graph Theory uses the spectrum of adjacency matrices associated to graphs in order to understand properties of the graph. In incidence geometry, we often encounter highly symmetric graphs for which the spectrum can be easily determined. In this chapter we introduce the necessary tools from spectral graph theory which we will later use in the construction of relevant graphs.

Definition 3.1. Let $G = (\{v_1, \dots, v_n\}, E)$ be a graph, its associated adjacency matrix is the matrix $A = (a_{i,j})_{1 \leq i,j \leq n} := (\mathbb{1}_{\{v_i, v_j\} \in E})_{1 \leq i,j \leq n}$. The spectrum of the graph are the n real eigenvalues of A , $\lambda_1 \leq \dots \leq \lambda_n$, which will always exist since A is symmetric.

Example 3.2. Consider the star graph on $n + 1$ vertices, S_{n+1} , a quick check gives that the adjacency matrix of S_{n+1} is the $(n + 1) \times (n + 1)$ matrix:

$$A(S_{n+1}) = \begin{bmatrix} 0 & 1 & 1 & \cdots & 1 \\ 1 & 0 & 0 & \cdots & 0 \\ 1 & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 0 & 0 & \cdots & 0 \end{bmatrix}$$

We can quickly verify that $\chi(S_{n+1}) = X^{n+1} - nX^n$ and thus the eigenvalues are 0 and n with multiplicity n and 1 respectively.

Definition 3.3. Let $d \in \mathbb{N}$ and $G = (V, E)$ a graph. G is said to be d -regular if for all $v \in V$, the degree of v is d .

Proposition 3.4. For any d -regular graph, $G = ([n], E)$, has largest eigenvalue d and is of multiplicity 1 if and only if G is connected.

Proof. Claim the all 1s vector is an eigenvector for d . Consider A the adjacency matrix of G , since G is d -regular the sum of any row of A is d , thus $(1 \ \dots \ 1)A = d \cdot (1 \ \dots \ 1)$. Now let $\lambda \in \text{Spec}(A)$ be an eigenvalue of A with eigenvector $x = (x_1 \ \dots \ x_n)$. Let $m \in [n]$ be the index such that $|x_m|$ is maximal. Then taking the m -th equation of the system $xA = \lambda x$ we find $|\lambda x_m| = \left| \sum_{i \in N(m)} x_i \right| \leq d|x_m|$, where $\#N(m) = d$ by d -regularity.

Suppose G is disconnected, let C be one of its connected components. C is a d -regular subgraph of G and thus has eigenvalue d with some eigen vector x_C , now choosing the vector $x = (x_C \ 0 \ \dots \ 0)$, we notice that in G , d has a new linearly independent eigenvector.

Suppose G is connected. We shall show that if $(x_1 \ \dots \ x_n)$ is an eigenvector of d , then $x_1 = \dots = x_n$. Let x_m be the largest of the coefficients, then we notice that $dx_m = \sum_{i \in N(v_m)} x_i \leq dx_m$, thus equality must hold for all $i \in N(v_m)$. Iterating this on these new vertices, by connectedness of G , we conclude that all the coefficients will be equal. $\square$

3.1. Cayley Graphs. A powerful way of generating graphs with a lot of symmetry is to consider the so called *Cayley graphs* of a group G .

Definition 3.5. Let G be a group and S a generating set of the group. Then the Cayley graph (possibly infinite) of G with respect to S is an edge colored, directed graph whose vertices are the elements of G and for every vertex $g \in G$ and generating element $s \in S$, there is a directed edge from g to gs with color c_s .

Example 3.6. The Cayley graph of the free group generated by two elements a, b with respect to the generating set $S = \{a, b\}$ has a fractal like structure.

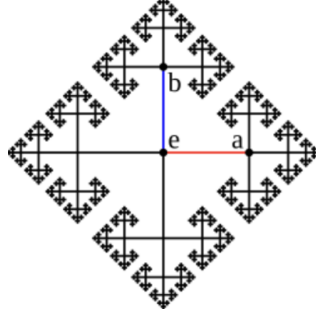


FIGURE 1. Free group Cayley graph

We state a basic result on the spectrum of the Cayley graphs of abelian groups and omit the proof which can be found in [7].

Theorem 3.7. *Let G be a finite abelian group, then the eigenvalues of its Cayley graph $\lambda_1, \dots, \lambda_{\#G}$ with respect to a generating set S are given by*

$$\lambda_j = \sum_{s \in S} e^{2\pi i j s / \#G}$$

3.2. Strongly regular graphs. *We now introduce the notion of strongly regular graphs, an extremely symmetric kind of graph which produces many pretty pictures! A powerful and amazing property of strongly regular graphs is that its spectrum is completely determined by the parameters of the graph, which makes them easy to work with.*

Definition 3.8. A graph G is said to be strongly regular with parameters (n, d, λ, μ) if it is d -regular any two neighboring vertices contain λ common neighbors and any two non-neighboring vertices contain μ common neighbors. If G is strongly regular with parameters (n, d, λ, μ) we shall say it is $\text{srg}(n, d, \lambda, \mu)$.

Example 3.9. The cycle graph of order 5, C_5 is strongly regular with parameters $(5, 2, 1, 1)$ or the Clebsch graph is a $\text{srg}(16, 5, 0, 2)$.

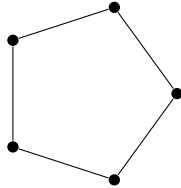
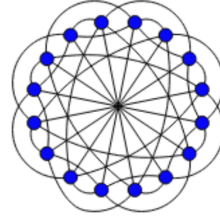
FIGURE 2. C_5 

FIGURE 3. Clebsch graph

Theorem 3.10. *Let $G = ([n], E)$ be a $\text{srg}(n, d, \lambda, \mu)$ with $d \geq \mu > 0$, then the eigen values of G are d, λ_1, λ_2 with multiplicities $1, m_1, m_2$ respectively where:*

$$\lambda_1, \lambda_2 = \frac{1}{2} \left(\lambda - \mu \pm \sqrt{\Delta} \right)$$

$$m_1, m_2 = \frac{1}{2} \left(n - 1 \mp \frac{2d + (n-1)(\lambda - \mu)}{\sqrt{\Delta}} \right)$$

Where the discriminant is given by $\Delta := (\lambda - \mu)^2 - 4(d - \mu)$.

Proof. Clearly G must be a connected graph since $\mu \geq 1$, thus by 3.4, we know that d is an eigenvalue with multiplicity 1. Let A be the adjacency matrix of G , it is well known each entry $\tilde{a}_{i,j}$ of A^2 is the number of length two paths from vertex i to j in G . For any vertex i , the number of length two paths from i to itself is d . If $i \neq j$ are joined by an edge, then $\tilde{a}_{i,j} = \lambda$ and if they aren't then $\tilde{a}_{i,j} = \mu$. Allowing us to conclude:

$$A^2 = \lambda A + dI_n + \mu(J - A - I_n)$$

Evaluating at an eigenvector x corresponding to some eigenvalue ν we find:

$$\nu^2 x + (\mu - \lambda)\nu x + (\mu - d)x + \mu Jx = 0$$

However, as x is orthogonal to the eigenvector associated to d , we have $Jx = 0$, we deduce from the quadratic equation in ν the two solutions for λ_1, λ_2 . To compute the multiplicities we can simply solve the system of equations

$$\begin{cases} 1 + m_1 + m_2 = n \\ \text{tr}(A) = d + m_1\lambda_1 + m_2\lambda_2 = 0 \end{cases}$$

□

3.3. Pseudorandom graphs. Lastly we introduce the pseudorandom graphs. These graphs act similar to graphs constructed randomly, this idea is captured in the following lemma. It intuitively states that the edges of a certain d -regular graph are evenly distributed throughout the graph.

Lemma 3.11 (Expander mixing lemma). *Let $G = (V, E)$ be a d -regular graph on n vertices with eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$ and $\lambda := \max(\lambda_2, -\lambda_n)$. Then for any two subsets $S, T \subset V$ of the vertices of G , denoting, $E(S, T) = \{\{s, t\} \in E : s \in S, t \in T\}$ we have:*

$$\left| \#E(S, T) - d \frac{\#S \cdot \#T}{n} \right| \leq \lambda \sqrt{\#S \cdot \#T}$$

Proof. Let $V = \{v_1, v_2, \dots, v_n\}$ and A be the adjacency matrix of G . Observe that $\#E(S, T) = {}^t \mathbb{1}_S A \mathbb{1}_T$, where ${}^t \mathbb{1}_X := (\mathbb{1}[v_1 \in X], \dots, \mathbb{1}[v_n \in X])$ is the indicator vector of $X \subset V$. Hence, $\left| \#E(S, T) - d \frac{\#S \cdot \#T}{n} \right| = |\mathbb{1}_S (A - \frac{d}{n} J) \mathbb{1}_T|$,

denoting $B = A - \frac{d}{n}J$ and noticing $|\mathbb{1}_S(A - \frac{d}{n}J)\mathbb{1}_T| = \langle \mathbb{1}_S, B\mathbb{1}_T \rangle$ we may apply Cauchy-Schwartz to find:

$$\left| \#E(S, T) - d \frac{\#S \cdot \#T}{n} \right| \leq \|\mathbb{1}_S\| \|B\mathbb{1}_T\| = \sqrt{\#S} \|B\mathbb{1}_T\|$$

Now simply noting that A and J commute, we may diagonalize B . However, note that for the maximal eigenvalue of A , which is d , the corresponding all 1s vector is common with J . Furthermore all the other eigenvalues of J are zero, we conclude that B has eigenvalues $0, \lambda_2, \dots, \lambda_n$, denote $(v_1, \dots, v_n)$ the corresponding eigenvectors. Thus, as B is symmetric:

$$\begin{aligned} \|B\mathbb{1}_T\|^2 &= \langle B\mathbb{1}_T, B\mathbb{1}_T \rangle \\ &= \langle \mathbb{1}_T, B^2\mathbb{1}_T \rangle = \langle \mathbb{1}_T, M^2 \sum_{i=1}^n \langle \mathbb{1}_T, v_i \rangle v_i \rangle = \sum_{i=2}^n \lambda_i^2 \langle \mathbb{1}_T, v_i \rangle^2 \leq \#T \lambda^2 \end{aligned}$$

Allowing us to conclude the proof. $\square$

Remark 3.12. The expander lemma says that given two arbitrary set of vertices on the graph, the number of edges between them is close to expected number of them when constructing the graph randomly. More specifically, where the random construction is such that an edge appears between two vertices with probability $\frac{d}{n}$.

We give a definition which extends on the notion of d -regularity adding a parameter λ which bounds the eigenvalues, so that the expander lemma may be easily stated on the graph.

Definition 3.13. We call a graph G a (n, d, λ) -graph if it is a d -regular graph on n vertices and such that all the eigenvalues except for the largest has absolute value no more than λ .

Corollary 3.14. For an (n, d, λ) -graph G we have $\alpha(G) \leq \frac{n\lambda}{d}$.³

Proof. Denote I the largest anti-clique in G , applying the expander lemma on I we find $|\#E(I, I) - d \frac{\#I^2}{n}| \leq \lambda \#I$ but noting $\#E(I, I) = 0$ we are able to conclude the result. $\square$

4. INCIDENCE GEOMETRY

Incidence geometry studies geometric properties which are independent of distance or angle. Instead we only consider the notion of *incidence*, what point is on which line? This is illustrated in the following definition.

Definition 4.1. An incidence geometry is a triplet of sets $(\mathcal{P}, \mathcal{B}, \mathcal{I})$, where $\mathcal{I} \subset \mathcal{P} \times \mathcal{B}$ such that for any $b \in \mathcal{B}$ there are at least two distinct $x, y \in \mathcal{P}$ with $(x, b), (y, b) \in \mathcal{I}$. The elements of $\mathcal{P}$ are called point, the elements of $\mathcal{B}$

³ $\alpha(G)$ traditionally denotes the cardinality of the largest set of vertices such that no two vertex is joint by an edge, called the independence number of G .

blocks and $\mathcal{I}$ is the incidence relation. We shall say a point p and a block b are incident, p lies on b or that b crosses through p whenever $(p, b) \in \mathcal{I}$.

Remark 4.2. Most often we will replace the notation $\mathcal{B}$ with $\mathcal{L}$ (or $\mathcal{C}$) and call their elements lines (resp. circles) instead of blocks. However, this is just a matter of geometric intuition. Furthermore, instead of writing $(p, b) \in \mathcal{I}$, we may use the notation $p \in b$. When not stated explicitly it will be typical to define $\mathcal{I}$ as $(p, l) \in \mathcal{I}$ if and only if $p \in l$.

Definition 4.1 reflects the above motivation, the only retained information about the geometry is the incidence between points and lines encoded in $\mathcal{I}$.⁴

Example 4.3. We can consider the set of points $\mathcal{P} = [4]$, lines $\mathcal{L} = \binom{[4]}{2}$ to be all subsets of size two and the incidence relation to be $\in$. This gives rise to the Affine plane of order 2.

Example 4.4. Consider an incidence geometry in the real plane $\mathbb{R}^2$, choose the set of points $\mathcal{P}$ to be $\mathbb{R}^2$, the set of lines $\mathcal{L}$ to contain all sets of the form $\{(x, mx + c) : x \in \mathbb{R}\}$ (slanted and horizontal lines) or $\{(c, x) : x \in \mathbb{R}\}$ (vertical lines), where $m, c \in \mathbb{R}$, and the incidence relation $\mathcal{I}$ to be defined as $(x, l) \in \mathcal{I}$ if and only if $x \in l$. Notice that every pair of distinct points $v, w \in \mathcal{P}$ defines a unique line in $\mathcal{L}$. Furthermore, for any point $v \in \mathcal{P}$ and line $l \in \mathcal{L}$ such that l doesn't cross v , there is a unique line $l' \in \mathcal{L}$ crossing v which is parallel to l . This example is called the Euclidean plane and we shall later see that it is a so-called *affine plane*.

Example 4.5. Let $(\mathcal{P}, \mathcal{C}, \mathcal{I})$ be a general incidence structure where we shall call the elements of $\mathcal{C}$ circles. This structure is called a Möbius plane or an inversive plane if any 3 distinct points $p, q, r \in \mathcal{P}$ defines a unique circle, for any circle $c \in \mathcal{C}$, points $p \in c$ and $q \notin c$, there is a unique circle $c_0 \in \mathcal{C}$ such that $p, q \in c_0$ and $c \cap c_0 = \{p\}$, as in figure 4, and there are at least four points not all on a single circle.

A particular case of this structure is the Real Inversive plane which can be defined as the point line geometry $(\mathcal{P}, \mathcal{C}, \mathcal{I})$ where $\mathcal{P} = \mathbb{S}^2$, the unit sphere in $\mathbb{R}^3$, and $\mathcal{C}$ to be the circles along the sphere. Equivalently, we may describe the geometry using complex numbers, the points $\mathcal{P} := \mathbb{C} \cup \{\infty\}$ are the complex numbers plus a point at infinity and the circles $\mathcal{C}$ are either passing through ∞ , parametrized as $\{z \in \mathbb{C} : az + \overline{a}z + b = 0\} \cup \{\infty\}$ where $a \in \mathbb{C}^*, b \in \mathbb{R}$ or are regular circles in $\mathbb{C}$, $\{z \in \mathbb{C} : (z - z_0)\overline{(z - z_0)} = d\}$, $z_0 \in \mathbb{C}, d \in \mathbb{R}_{>0}$. This description of the real inversive plane allows us to consider natural maps on the points of $\mathcal{P}$ and how they act on $\mathcal{C}$. For example $\mathcal{C}$ is stable under the following maps: $\mathcal{P} \ni z \mapsto rz$ (rotation + dilatation by r), $\mathcal{P} \ni z \mapsto z + r$

⁴The notion of an incidence geometry is actually fully analogous to that of hypergraphs. Given an incidence geometry $G = (\mathcal{P}, \mathcal{B}, \mathcal{I})$ we can define the hypergraph $H = (\mathcal{P}, E)$ where $E := \{\{p \in \mathcal{P} : (p, b) \in \mathcal{I}\} : b \in \mathcal{B}\}$. Similarly given a hypergraph $H = (V, E)$ we can define a corresponding geometry G .

(translation by r) where both dilatation and translation maps $\infty \rightarrow \infty$ and lastly $\mathbb{C} \ni z \mapsto \frac{1}{z}$ (inversion) where $0 \rightarrow \infty, \infty \rightarrow 0$.

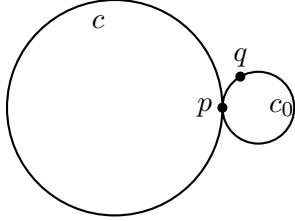


FIGURE 4. Möbius plane

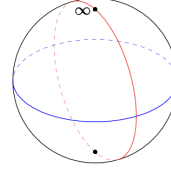


FIGURE 5. Real Inversive plane

Let us provide some more definitions which attempt to highlight interesting properties of geometries.

Definition 4.6. Let $G = (\mathcal{P}, \mathcal{L}, \mathcal{I})$ be an incidence geometry. Then G is said to be a partial linear space if for each pair of distinct points $x, y \in \mathcal{P}$ there is at most one line $l \in \mathcal{L}$ incident to both x and y . Furthermore, if for each pair of distinct points there is exactly one such line, then we say G is a linear space. In the case of a linear space we denote the unique line defined by the two points $x, y \in \mathcal{P}$ by $xy \in \mathcal{L}$.

The familiar Euclidean plane 4.4 is an example of a linear space, since every two points defines a unique line. The affine plane of order 2, $\mathbb{A}_2$ is also a linear space, as can be seen in figure ??, since every pair of points x, y is contained in the line $\{x, y\} \in \mathcal{L}$.

Definition 4.7. Let $G_1 = (\mathcal{P}_1, \mathcal{L}_1, \mathcal{I}_1), G_2 = (\mathcal{P}_2, \mathcal{L}_2, \mathcal{I}_2)$ be two incidence geometries. We say a pair of bijections $f_{\mathcal{P}} : \mathcal{P}_1 \rightarrow \mathcal{P}_2, f_{\mathcal{L}} : \mathcal{L}_1 \rightarrow \mathcal{L}_2$ is an isomorphism if for any $(p, l) \in \mathcal{I}_1$ we have $(f_{\mathcal{P}}(p), f_{\mathcal{L}}(l)) \in \mathcal{I}_2$.

This allows to describe geometries which have the same structure and are equivalent up to a change of labels. Note that for linear spaces, defining the function $f_{\mathcal{P}}$ is actually sufficient, since $f_{\mathcal{L}}$ will be fully determined by $f_{\mathcal{P}}$.

Definition 4.8. The automorphism group of a point line geometry G , denoted $\text{Aut}(G)$, is the group of isomorphisms from G into itself.

Example 4.9. Consider the automorphism group of the point line geometry, S_n , induced by the star graph on n vertices. It is not to difficult to observe that $\text{Aut}(S_n)$ is isomorphic to the symmetric group on $n - 1$ elements since automorphisms of the graph must fix the central node and may permute the leaf nodes in any order.

Definition 4.10. Let $G = (\mathcal{P}, \mathcal{L}, \mathcal{I})$ be an incidence geometry, we define the dual of G , denoted $G^* = (\mathcal{P}, \mathcal{L}, \mathcal{I}^*)$ to be the incidence structure where we

have inverted the points and lines in G and $\mathcal{I}^* := \{(l, p) : (p, l) \in \mathcal{I}\} \subset \mathcal{L} \times \mathcal{P}$. Furthermore, G is said to be self dual if it is isomorphic to its dual.

Lastly, the point of interest for us about these geometries are the graphs they produce, we define the incidence graph associated to an incidence geometry G .

Definition 4.11. Let $G = (\mathcal{P}, \mathcal{L}, \mathcal{I})$ be an incidence geometry. The incidence graph of G , is the graph whose vertices are the points and lines of G and where a point p is joined with a line l if and only if the two are incident in the geometry G , we denote this joining edge by pl .

4.1. Affine Planes. An affine plane is a linear incidence structure which captures the incidence of the typical Euclidean plane.

Definition 4.12. An incidence geometry $G = (\mathcal{P}, \mathcal{L}, \mathcal{I})$ is said to be an affine plane if it is a linear space and the following hold:

- For each point p and line l not crossing through p , there is a unique line l' such that $l \cap l' = \emptyset$ ⁵
- There are four points no three of which are collinear

Example 4.13. As we have mentioned above, ?? and 4.4 are both affine planes, one finite and the other infinite.

Example 4.14. The Moulton Plane $\mathcal{M} = (\mathbb{R}^2, \mathcal{L}, \mathcal{I})$ over $\mathbb{R}^2$ is the affine plane whose lines are either vertical lines $\{(c, x) : x \in \mathbb{R}\}$ or perturbed sloped lines written as $\{(x, m \circ x + c) : x \in \mathbb{R}\}$ where the binary operation $\circ$ is defined as

$$x \circ m = \begin{cases} mx & \text{if } m \text{ or } x \leq 0, \\ 2mx & \text{if } m \text{ and } x > 0. \end{cases}$$

The Moulton plane is the simplest affine plane where Desargues's theorem does not hold. Such planes are called *non-Desarguesian* or *non-classical*.

Definition 4.15. Let $\mathcal{A}$ be an affine plane and define the *parallelism* relation $\parallel$ on the set of lines $\mathcal{L}$ by $l \parallel l' \iff l \cap l' = \emptyset$ or $l = l'$. Parallelism is an equivalence relation on $\mathcal{L}$ and its equivalence classes are called parallel classes.

In the case of finite affine planes, affine planes can be categorized by a notion of order which is given in the following theorem.

Theorem 4.16. Let $\mathcal{A} = (\mathcal{P}, \mathcal{L}, \mathcal{I})$ be a finite affine plane, then $\exists n \in \mathbb{N}_{\geq 2}$, called the order of the plane, such that:

- (1) There are n^2 points and $n^2 + n$ lines.
- (2) Through each point crosses $n + 1$ lines
- (3) Each line contains n points
- (4) There are $n + 1$ parallel classes each containing n lines

⁵The notation $l \cap l'$ designates the set of points common to both l and l' .

Proof. By definition of $\mathcal{A}$, there exists 4 points $x, a, b, c \in \mathcal{P}$ no three of which are collinear, denote $n + 1$ the number of lines crossing through x . Observe that since the lines $xa, xb, xc \in \mathcal{L}$ are all distinct and crossing through x we have $n + 1 \geq 3$, claim that n satisfies all the above.

(2): Let $p \in \mathcal{P} - \{x\}$ be an arbitrary point and denote $\mathcal{L}_p$, (resp. $\mathcal{L}_x$) the set of lines crossing through p (resp. x). For each line $l \in \mathcal{L}_x$ there is a unique line $l' \in \mathcal{L}_p$ parallel to l ⁶, define thus the function $f : \mathcal{L}_x \rightarrow \mathcal{L}_p$ $f(l) = l'$ as above. By a symmetric argument on $\mathcal{L}_p$ we conclude f must actually be bijective.

(3): Let $l \in \mathcal{L}$ and consider a point p not on l , we write $\mathcal{P}_l$ the set of points on l and l' the unique line parallel to l crossing through p . Let $f : \mathcal{P}_l \rightarrow \mathcal{L}_p - \{l'\}$ where for all $q \in \mathcal{P}_l$, $f(q) := pq$ is the unique line crossing through both p and q . Inversely, $\forall m \in \mathcal{L}_p - \{l'\}$ there is a unique point q on $l \cap m$, the function $f^{-1} : \mathcal{L}_p - \{l'\} \rightarrow \mathcal{P}_l$ with $f(m) := q$ is the inverse of f . We conclude f is bijective and $\#\mathcal{P}_l = \#\mathcal{L}_p - 1 = n$.

(4): Let $p \in \mathcal{P}$ be an arbitrary point and let $S = \{[l_1], [l_2], \dots, [l_m]\}$ be the set of distinct parallel classes. Since none of the lines crossing through p are parallel to each other, we find that $m \geq n + 1$. If it were strictly larger, let $[l]$ be the additional parallel class, with $[l] \cap \mathcal{L}_p = \emptyset$, then since $x \notin l$ there is a line l^* crossing through x parallel to l , a contradiction since $[l] \cap \mathcal{L}_p = \emptyset$. Furthermore, if l and m are two lines crossing through p , let $f : \mathcal{P}_m \rightarrow [l]$ such that for all $q \in \mathcal{P}_m$ $f(q) := l_q$ where l_q is the unique line crossing through q and parallel to l . The inverse exists since given $l^* \in [l]$ there is unique point on $m \cap l^*$ which is exactly $f^{-1}(\{l^*\})$.

(1): For a parallel class C , $\mathcal{P} = \bigsqcup_{l \in C} l$, thus the number of points by (3) and (4) is n^2 . Since $\mathcal{L} = \bigsqcup_{C \text{ parallel class}} C$ we find by (4) that the number of lines is $n^2 + n$. $\square$

Let us present some fundamental examples and constructions of finite affine planes.

Definition 4.17. We define the classical affine plane of order q as the point line geometry $\mathbb{A}_q = (\mathbb{F}_q^2, \mathcal{L}, \mathcal{I})$, where lines are either of the form $\{(x, mx + c) : x \in \mathbb{F}_q\}$ or of the form $\{(c, x) : x \in \mathbb{F}_q\}$ for $m, c \in \mathbb{F}_q$ and $\mathcal{I}$ is defined as the $\in$ inclusion.

Remark 4.18. Equivalently, this may be formulated as the point line geometry where points are given by a 2 dimensional $\mathbb{F}_q$ -vector space V and lines by all the translations of $L := \{1 \text{ dimensional subspaces of } V\}$, $\mathcal{L} := \bigcup_{x \in V} L + x$. We may actually generalize this observation in the following definition to vector spaces of any dimension.

Definition 4.19. Let V be a $2n$ dimensional $\mathbb{F}_q$ -vector space. A *spread* $\Sigma = \{V_0, \dots, V_m\}$ is a set of n dimensional subspaces of V such that $\bigsqcup_{U \in \Sigma} U =$

⁶More precisely, we are applying the first axiom of affine planes where in the case in which $p \in l$ we pick $l' = l$.

$\{0\} = V - \{0\}$. Then the affine plane associated to the spread Σ , $\mathcal{A}_\Sigma$, has its points in V and the lines are cosets of $U \in \Sigma$.⁷

Notice that for any spread Σ :

$$\#\Sigma = \frac{\#V - 1}{(\#V_1 - 1)} = \frac{q^{2n} - 1}{q^n - 1} = q^n + 1$$

Furthermore, for any distinct components, $V_i, V_j \in \Sigma$, $V_i \oplus V_j = V$.

Proposition 4.20. *Let V be $2n$ dimensional $\mathbb{F}_q$ -vector space and $\Sigma = \{V_0, \dots, V_{q^n}\}$ a spread, then $\mathcal{A}_\Sigma$ is an affine plane.*

Proof. Let $v, w \in V$, there exists a unique $V_i \in \Sigma$ such that $v - w \in V_i$, but it is then clear that both v and w are contained in the coset $V_i + w$. If for some $j \in [q^n + 1]$, $x \in V$, we have $v, w \in V_j + x$, then $v - x, w - x \in V_j$ and $v - w \in V_j$ which implies $V_j = V_i$ but since the family $(V_i + x)_{x \in V}$ is pairwise disjoint, this implies $x = w$. Let $V_i + y \in \mathcal{L}$ and $x \in V$ not on $V_i + y$ be given, then $V_i + x$ is the unique line parallel to $V_i + y$ containing x . We have $q^n + 1 \geq 3$, let x, y be non-zero points on two distinct lines, then the family $0, x, y, x + y$ has no three points collinear. $\square$

A systematic way of constructing spread is using so called slope matrices which are used as follows.

Proposition 4.21. *Let $(M_i)_{i=0}^{q^n-1} \in M_n(\mathbb{F}_q)$ be a family of $n \times n$ matrices with coefficients in $\mathbb{F}_q$ such that for any pair $i \neq j$, $M_i - M_j$ is an invertible matrix. Then the set $\Sigma = \{V_i : i = 0, \dots, q^n\}$ where for $i = 0, \dots, q^n - 1$, $V_i := \{(x, xM_i) : X \in \mathbb{F}_q^n\}$ and $V_{q^n} = \{(0, x) : X \in \mathbb{F}_q^n\}$ defines a spread over $\mathbb{F}_q^{2n}$.*

Proof. Well clearly all $\dim_{\mathbb{F}_q}(V_i) = n$. Let a pair $i, j \in \{0, 1, \dots, q^n - 1\}$, $i \neq j$ and $v \in V_i \cap V_j$ be given. Then, $v = (x, xM_i) = (x, xM_j)$ and $(0, x(M_i - M_j)) = 0$ by invertibility of $M_i - M_j$ we conclude $v = 0$. In the case in which i or j is q^n the same holds as we easily find $x = 0$. Furthermore, since $\#\Sigma = q^n + 1$ and each $V_i \cap V_j = \{0\}$, we conclude $\bigsqcup_{U \in \Sigma} U - \{0\} = \mathbb{F}_q^{2n} - \{0\}$. $\square$

Example 4.22. The following set of slope matrices over $\mathbb{F}_3$ gives rise to the Hall Plane of order 3, which one can prove is not isomorphic to the classical plane of order 3.⁸

$$\begin{aligned} & \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}, \quad \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}, \quad \begin{bmatrix} 2 & 2 \\ 2 & 1 \end{bmatrix}, \\ & \begin{bmatrix} 0 & 1 \\ 2 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & 2 \\ 1 & 0 \end{bmatrix}, \quad \begin{bmatrix} 1 & 2 \\ 2 & 2 \end{bmatrix}, \quad \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \end{aligned}$$

⁷Choosing $n = 1$ gives precisely the definition of classical affine planes, however, for larger n spreads will produce constructions of non-classical or so called non-Desarguesian planes.

⁸One may prove this by illustrating that the plane does not verify Desargues's theorem or Pappus's theorem.

We have now considered a variety of affine planes, classical and non-classical, none the less, it is still unknown whether all finite affine planes have a prime power order.

Conjecture 4.23. $m \in \mathbb{N}$ is the order of an affine plane if and only if m is a prime power.

TABLE 2. Number of order n affine planes

n	2,3,4,5	6	7,8	9	10	11	12	13	14	16
# of order n planes	1	0	1	7	0	≥ 1	≥ 0	≥ 1	0	≥ 88
n	17	19	23	25		27	29	31	37	
# of order n planes	≥ 1	≥ 1	≥ 1	≥ 2959		≥ 31	≥ 1	≥ 1	≥ 1	≥ 1

Conjecture 4.24. The only affine plane of prime order p is the classical affine plane $\mathbb{A}_p$.

As is illustrated in 2, no non-Desarguesian affine plane has been found for prime orders.

4.2. Projective Planes. Projective planes extends the notion of affine planes in an attempt to eliminate parallel lines. A consequence is that any two lines in the geometry intersects at a unique point, which, as we can imagine, is a property that can simplify many ideas and proofs. To construct such planes we may think about adding a line at infinity and force parallel lines to intersect at this new line. We first provide an axiomatic definition of projective planes.

Definition 4.25. Let $G = (\mathcal{P}, \mathcal{L}, \mathcal{I})$ be an incidence geometry, G is said to be a projective plane if it is a linear space and the following hold:

- Any two distinct lines $l, l' \in \mathcal{L}$ intersect at a unique point.
- There exists 4 points no three of which are collinear

Example 4.26. The smallest example of a projective plane, composed of 7 points and 7 lines, is the classical projective plane of order two $PG(2, 2)$ also called the Fano plane. As in 6, each pair of lines intersect at a unique point and the construction from the affine plane is shown in 7.

Let us more generally construct the classical projective plane of order q , denoted $PG(q, 2)$. Let $\mathbb{A}_q = (\mathcal{P}, \mathcal{L}, \mathcal{I})$ be the affine plane of order q , we define the new line and points at infinity, $l_\infty = \{p_0, \dots, p_q\}$. For every parallel class $C_i \in \{C_0, C_1, \dots, C_q\}$ and for every $l \in C_i$ we add to $\mathcal{I}$ the relation (p_i, l) . This new incidence structure $PG(q, 2)$ is a projective plane. It is quite easy to see that the above construction generalizes to any affine plane and may be performed backwards.⁹

⁹Arbitrarily pick a line l in the projective plane π and remove it along with all its points, the resulting geometry is an affine plane.

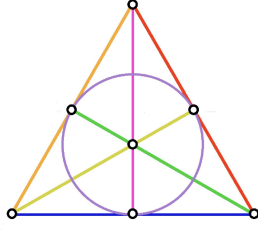
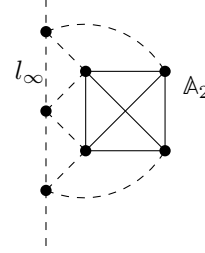


FIGURE 6. Fano Plane

FIGURE 7. $\mathbb{A}_2$ Extension

With this correspondence between projective and affine planes we may translate some theorems to projective planes.

Theorem 4.27. *Let $\pi = (\mathcal{P}, \mathcal{L}, \mathcal{I})$ be a projective plane. Then there exists $n \in \mathbb{N}_{\geq 2}$ called the order of π such that the following are true:*

- (1) $\#\mathcal{P} = \#\mathcal{L} = n^2 + n + 1$
- (2) *Through each point crosses $n + 1$ lines*
- (3) *Each line contains $n + 1$ points*

Proof. Let $l_\infty \in \mathcal{L}$ be an arbitrary line of π and consider $\mathcal{A}$ the affine plane induced by removing l_∞ from π . By 4.16, $\exists n \in \mathbb{N}_{\geq 2}$ such that 1 2 3 4 are all true. Since we have removed l_∞ which contained $n + 1$ points we deduce (1). Furthermore, in $\mathcal{A}$ every line contains n points and through every point crosses $n + 1$ lines. As we extend each line to l_∞ adding an additional point and not altering points already in $\mathcal{A}$ we conclude that (2) and (3) is true for any point and line in $\mathcal{A}$. Now for any point $p \in l_\infty$, since there are n lines in each parallel class of $\mathcal{A}$ we deduce that $n + 1$ lines pass through p . Lastly, as there are $n + 1$ parallel classes, l_∞ contains $n + 1$ points, concluding the proof. $\square$

Actually, the symmetry captured by 4.27 arises from the stability of projective planes by duality.

Proposition 4.28. *Let π be a projective plane, then its dual π^* , is a projective plane as well.*

Proof. Let p, p' be two lines of π^* . By linearity of the space, there is unique line l of π crossing through p and p' and thus in π^* , $p \cap p' = \{l\}$. Furthermore, since there are 4 points a, b, c, d in π no three of which are collinear, then the lines ab, bc, cd, da satisfy the same property in π^* . If there were a line p of π^* containing say ab, bc, cd , then ab and bc would intersect at both b and p in π , implying $b = p$ and $b, c, d \in cd$, a contradiction. $\square$

Remark 4.29. Using this observation on the dual of a projective plane, we could have simplified the proof of 4.27 by proving the statements only for the points of the plane and concluding the same properties must hold on the lines by considering the dual. This phenomenon is a more general fact, if a collection of incidence geometries satisfy a certain property, then the

dual property will be true for their dual geometries, in the case of projective planes this implies new properties on the same collection of objects.

An equivalent way of constructing the classical projective plane is to consider a 3 dimensional vector space V over $\mathbb{F}_q$ and let the points (resp. lines) of the geometry be the 1 dimensional (resp 2 dimensional) subspaces of V with the incidence relation to be inclusion $\subset$. This geometry defines a projective plane since any two planes π_1, π_2 intersect at a 1 dimensional line of V , we shall show that this structure is isomorphic to $PG(q, 2)$.

Theorem 4.30. *The projective plane π whose points are 1 dimensional subspaces of $\mathbb{F}_q^3$ and lines 2 dimensional subspaces of $\mathbb{F}_q^3$ is isomorphic to $PG(q, 2)$.*

Proof. Let $V = \mathbb{F}_q^3$, $\mathcal{A} = (\mathcal{P}_{\mathcal{A}}, \mathcal{L}_{\mathcal{A}}, \mathcal{I}_{\mathcal{A}}) := \{(x, y, 1) \in V\}$ the classical affine plane of order q embedded in V and $\pi = (\mathcal{P}_{\pi}, \mathcal{L}_{\pi}, \mathcal{I}_{\pi})$ the projective plane as described above. Define the map $f : \mathcal{P}_{\mathcal{A}} \rightarrow \mathcal{P}_{\pi}$ where a point in $\mathcal{A}$ is mapped to the unique 1 dimensional line crossing through the point. Furthermore, the map is injective since 1 dimensional lines will intersect $\mathcal{A}$ at a single point. Thus, π has a sub-geometry isomorphic to the classical affine plane of order q , namely $f(\mathcal{A})$. Noting the plane $l_{\infty} := \{(x, y, 0) \in V\}$ as the line at infinity, we observe that π is the extension of $f(\mathcal{A})$ by l_{∞} . $\square$

Remark 4.31. From now on when we consider the projective plane of order q , we will adopt the point of view described above, which will turn to be a lot more approachable.

Corollary 4.32. *The classical projective plane of order q is self dual.*

Proof. Consider the classical projective plane $\pi = (\mathcal{P}, \mathcal{L}, \mathcal{I})$ of order q , we equip $\mathbb{F}_q^3$ with the standard inner-product. Then the map $\perp : \mathcal{P} \rightarrow \mathcal{L}$ which maps a 1 dimensional space to its perpendicular is an isomorphism. $\square$

This new construction of classical projective planes allows us to consider projective spaces of any dimension $n \geq 1$, denoted $PG(n, q)$.

Definition 4.33. The projective space of dimension n over a base field k , denoted $PG(n, k)$, is defined as a $n + 1$ tuple $(\mathcal{P}, \mathcal{L}, \Pi, \dots, \mathcal{H}, \mathcal{I})$ where $\mathcal{P}$ are the 1 dimensional lines of the vector space k^{n+1} , $\mathcal{L}$ the two dimensional planes of k^{n+1} , $\dots$, $\mathcal{H}$ the hyperplanes of k^{n+1} and the incidence relation is inclusion of spaces $\subset$. In the case where k is the finite field of order q , we denote the projective space simply by $PG(n, q)$.

Remark 4.34. A natural question to ask is can we give a closed form for the automorphism group $PG(n, q)$? One may guess that $\text{Aut}(PG(n, q))$ must be closely related to $\text{GL}_n(\mathbb{F}_q)$, since $\text{GL}_{n+1}(\mathbb{F}_q)$ acts linearly on $\mathbb{F}_q^{n+1}$, this induces an action of $\text{PGL}_n(\mathbb{F}_q) = \text{GL}_{n+1}(\mathbb{F}_q) / \langle I_n \rangle$ on $PG(n, q)$. However,

more than this, fixing a basis $(e_i)_{i=1}^n$ of $\mathbb{F}_q^{n+1}$, any automorphism σ of $\mathbb{F}_q^{n+1}$ induces an automorphism of $PG(n, q)$ via the map $f : \mathbb{F}_q^{n+1} \rightarrow \mathbb{F}_q^{n+1}$ where $\sum_{i=1}^n x_i e_i \mapsto \sum_{i=1}^n \sigma(x_i) e_i$. It turns out that such maps are the only type of automorphism of $PG(n, q)$ and $\text{Aut}(PG(n, q)) \cong \text{P}\Gamma\text{L}_{n+1}(\mathbb{F}_q)$, the set of semi linear automorphism¹⁰ of $\mathbb{F}_q^{n+1}$ quotiented by the multiples of identity. This is known as the fundamental theorem of projective geometry which is proved in [3].

With all this in place we may now consider points of classical projective planes more concretely as coordinates of the underlying k -vector space. Since a point, p , in $PG(n, k)$ is a 1 dimensional subspace of k^{n+1} , it may be expressed as the set $\{\lambda v : \lambda \in k\}$ or the equivalence class of v in $\sim$ where $v \sim w$ if and only if $v = \lambda w$. Fixing a basis $\mathcal{B}$ of k^{n+1} , we may express p as $p = \lambda(v_0, \dots, v_n)$ for any $\lambda \in k$. Furthermore, for convenience and uniqueness of the coordinates we may ensure the first non-zero coefficient is normalized to 1.

4.3. Conics and Ovals. We now consider certain curves in $PG(2, k)$ defined through the coordinate system of the plane which will allow us to construct extremal sets of points no three of which are collinear, called ovals and hyper ovals.

Definition 4.35. Let $\pi = (\mathcal{P}, \mathcal{L}, \mathcal{I})$ be the classical projective plane over a field k , $C \subset \mathcal{P}$ is said to be a conic of π if there exists a homogeneous polynomial $P \in k[x, y, z]$ of degree 2 such that $C = \{(p_1, p_2, p_3) \in \mathcal{P} : P(p_1, p_2, p_3) = 0\}$. Furthermore, the conic is said to be irreducible if the underlying polynomial P cannot be written as the product of two degree one polynomials with coefficients in k or any field extension of k . A conic not satisfying this property is called degenerate.

Example 4.36. The conic in $PG(2, \mathbb{R})$ defined by the polynomial $P(x, y, z) = x^2 + y^2$ is degenerate as $P(x, y, z) = (x - iy)(x + iy)$, where i is the imaginary unit. This conic C contains only the point $(0, 0, 1)$ of the projective plane.

Example 4.37. Consider the conic in $PG(2, \mathbb{R})$ defined by the polynomial $P(x, y, z) = x^2 + y^2 + z^2$. The conic defined by P is irreducible. Interestingly, we find that this conic is actually empty since its only solution in $\mathbb{R}$ are $x = y = z = 0$, which is not an element of $PG(2, \mathbb{R})$.

Example 4.38. The conic in $PG(2, k)$ defined by the polynomial $Q = y^2 - zx$ is irreducible and may be written parametrically as the set $C = \{(1, \lambda, \lambda^2) : \lambda \in k\} \cup \{(0, 0, 1)\}$.

Proposition 4.39. Let C be a conic and l a line of $PG(2, k)$, then either l intersects C at no more than 2 points or is fully contained in C .

¹⁰ $f : V \rightarrow V$ is a semilinear transformation of V if $\exists \sigma \in \text{Aut}(k)$ an automorphism of the base field k such that for any $v, w \in V, \lambda \in k$ $f(v + \lambda w) = f(v) + \sigma(\lambda)f(w)$.

Proof. We complete a basis of the 2 dimensional space l to a basis of k^3 . Now any point $p \in l$ has normalized coordinates $(0, 1, \lambda)$ for some $\lambda \in k$. Denote $Q(x, y, z) = ax^2 + by^2 + cz^2 + dxy + exz + fyz$ the polynomial associated to C . If a point $p \in l$ is also in C then $Q(0, 1, \lambda) = 0 \iff c\lambda^2 + f\lambda + b = 0$. Thus, either the quadratic in λ is non-zero and there are at most two solutions or $c = f = b = 0$ and the entirety of $l \subset C$. $\square$

If $p, q = (x_0, y_0, z_0), (x_1, y_1, z_1)$ are points in $PG(2, k)$ and C a conic in the plane such that $q \in C$ and $p \notin C$, then for any point $\lambda p + q, \lambda \in k$ on the line pq we find:

$$\begin{aligned} P(\lambda p + q) &= (2ax_0x_1 + 2by_0y_1 + 2cz_0z_1 + dx_0y_1 \\ &\quad + dx_1y_0 + ex_0z_1 + ex_1z_0 + fy_0z_1 + fy_1z_0)\lambda \\ &\quad + P(p)\lambda^2 \end{aligned}$$

Where we have taken P the polynomial defining C and the coefficients as above. Therefore, the solutions to $P(\lambda p + q) = 0$ are $\lambda = 0, -\frac{r}{P(p)}$, with r the coefficient of λ as above.

The equation has a unique solution if and only if $r = 0$, thus, if not all the $\frac{\partial P}{\partial x}(q) := 2ax_1 + dy_1 + ez_1, \frac{\partial P}{\partial y}(q) := dx_1 + 2by_1 + fz_1, \frac{\partial P}{\partial z}(q) := ex_1 + fy_1 + 2cz_1$ are zero then, the line given by:

$$\frac{\partial P}{\partial x}(q)x + \frac{\partial P}{\partial y}(q)y + \frac{\partial P}{\partial z}(q)z = 0$$

is the unique line intersecting C only at q , called the tangent of C at q .

Definition 4.40. Let C be a conic in $PG(2, k)$ defined by the polynomial $P \in k[x, y, z]$, then $q \in C$ is said to be a singular point if it annihilates all the partial derivatives of P . If C contains no singular points it is said to be non-singular.

By the above, we notice that if q is a singular point then there are infinitely many lines tangent to C at q . This sense of degeneracy at singular points motivates the following theorem.

Theorem 4.41. A conic C in $PG(n, k)$ is irreducible if and only if it is non-singular.

Proof. We only give the proof that an irreducible conic is non-singular, which is all we will use in the following, the converse is proved in [4]. Let C be a conic associated with $P \in k[x, y, z]$, $P(x, y, z) = ax^2 + by^2 + cz^2 + dxy + exz + fyz$ and suppose it is singular. Then $\exists q \in C$ such that all the partial derivatives of P cancel at q . We may fix a basis of the projective plane such that $q = (1, 0, 0)$ and thus we obtain $a = d = e = 0$ and thus over an appropriate

extension of k such that $b, f^2 - 4bc$ admit square roots:

$$\begin{aligned} P(x, y, z) &= by^2 + cz^2 + fyz \\ &= \left(\sqrt{b}y + \frac{f + \sqrt{f^2 - 4bc}}{2\sqrt{b}}z \right) \left(\sqrt{b}y + \frac{f - \sqrt{f^2 - 4bc}}{2\sqrt{b}}z \right) \end{aligned}$$

□

With this we have developed a tool to quickly check for irreducibility of conics by looking for singular points on the conic, a useful tool.

Corollary 4.42. *An irreducible conic C in $PG(2, q)$ which contains a point must contain exactly $q + 1$ points.*

Proof. Let x be a point on C , by irreducibility of C , x must be a non-singular point, therefore there is a unique line l tangent to C at x . By uniqueness of l , each of the other q lines crossing through x intersect C at a second point of the conic. Conversely, for each point p on the conic different from x , there is a unique line crossing through p and x . Thus, since there are $q + 1$ lines crossing through x , we conclude C contains $q + 1$ points. □

Definition 4.43. A subset S of n points in the classical projective plane over $\mathbb{F}_q$ is said to be an n -arc if no three points are collinear.

Example 4.44. Any non-empty irreducible conic C is a $q + 1$ -arc since a line will intersect C at less than 3 points. We note that an irreducible conic over $\mathbb{F}_q$, $q > 2$ cannot contain an entire line l . If it did then after an appropriate choice of basis, $l = \{(0, 1, \lambda) : \lambda \in k\}$, which allows us to deduce that $\forall \lambda \in k, c^2\lambda^2 + f\lambda + b^2 = 0$. Thus, we must have $b = c = f = 0$ and $P(x, y, z) = x(ax + dy + ez)$.

Proposition 4.45. *An n -arc S in $PG(2, q)$ can have at most $q + 2$ points.*

Proof. Let $x \in S$ and denote $\mathcal{L}_x$ the lines crossing through x , then the map $f : S - \{x\} \rightarrow \mathcal{L}_x$ which takes a point $s \in S - \{x\}$ to the unique line crossing through s and x is injective since no three points in S are collinear. We deduce that $\#S - \{x\} \leq \#\mathcal{L}_x = q - 1$ and conclude the proof. □

Remark 4.46. Equality can only hold if q is even. Suppose an arc S contains $q + 2$ points, then every line in the plane will intersect S at either 0 or 2 points. If there were a line l intersecting S at only one point, then picking $y \in l$ and not in S gives the same injection from S into the set of lines crossing through y , allowing us to conclude $\#S \leq q + 1$. Picking an arbitrary point not in S and considering the partition formed by the lines crossing through the point will pair up points in S implying q is even.

This extremal property of non-collinear points will be crucial in the explicit construction of graphs leading to bounds to $R(s, t)$, motivating us to define the following.

Definition 4.47. An oval of $PG(2, q)$ is a set of $q + 1$ points in $PG(2, q)$ no three of which are collinear. Similarly, a hyperoval of $PG(2, q)$ is a set of $q + 2$ points in $PG(2, q)$ no three of which are collinear.

4.4. Generalized Quadrangles. Lastly, we will introduce our first partial linear space, generalized quadrangles. These geometries will aim to construct *triangle free* geometries, which will be crucial in later constructions. Generalized quadrangles inherits its name as they are actually generalized n -gons for $n = 4$, which, intuitively, are partial linear spaces which don't contain m -gons for $2 \leq m < n$ but do contain n -gons. Furthermore, it is nice to note that for $n = 3$ we obtain projective planes, hence, generalized polygons encompass many of the structures we have considered.

Definition 4.48. A partial linear space G is said to be a generalized quadrangle if the following hold:

- For any point p and line l not containing p , there is a unique point $q \in l$ such that the line pq is in the geometry
- Each point is incident with at least two lines.

Proposition 4.49. *The dual of a generalized quadrangle remains a generalized quadrangle*

Proof. Let G be a generalized quadrangle and denote G^* its dual structure. Then for any point l of the dual and line p not containing l , we know in G that there is a unique point $p' \in l$ such that p and p' are on a common line l' . Thus in the dual, $l' \in p$ is the unique point such that l and l' are collinear. Furthermore, the second point follows from the definition of an incidence geometry. $\square$

Example 4.50. A *degenerate* example of a generalized quadrangle is an $n \times m$ grid. Let the set of points $\mathcal{P}$ be all tuples $(x, y) \in [n] \times [m]$ and the lines either horizontal $\{(x, c) : x \in [n]\}$ or vertical $\{(c, y) : y \in [m]\}$. Furthermore, the dual of the $n \times m$ grid is the complete bipartite graph $K_{n,m}$, a non-isomorphic generalized quadrangle. Observe that in these examples, there are two *types* of lines, vertical lines crossing through n points and horizontal lines through m points. Therefore, there is no notion of order as introduced in affine and projective geometries. However, it is known that other than these degenerate cases, generalized quadrangles may be categorized by order.

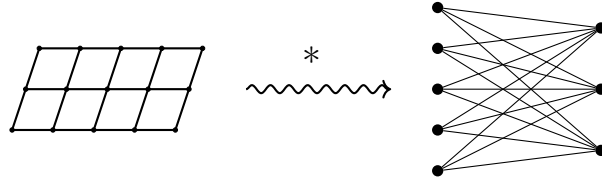


FIGURE 8. $n \times m$ grid and its dual $K_{n,m}$

Definition 4.51. An incidence geometry G is said to have order (s, t) if every line crosses through $s + 1$ points and every point of the geometry is incident to $t + 1$ lines.

Example 4.52. A generalized quadrangle of order $(2, 2)$ is the so called Doily, $\mathcal{D} = (\mathcal{P}, \mathcal{L}, \mathcal{I})$. Consider the geometry where $\mathcal{P}$ is the set of partitions of $[6]$ into parts of size 2 and two points are on a same line if the corresponding partitions contain a common part. Suppose $x \in \mathcal{P}$ and $l \in \mathcal{L}$ is a line not containing x , then all points of l have a common part $P = \{a, b\}$ in their partition. Since $x \notin l$, we have $P \notin x$, thus, $x = \{P_a, P_b, P_0\}$ will have two distinct parts P_a, P_b one containing a and the other b . Observing that the only possible common part between x and a point $p \in l$ is P_0 , we conclude that the unique point on l collinear to x must be the partition $\{\{a, b\}, P_0, [6] - \{a, b\} \cup P_0\}$.

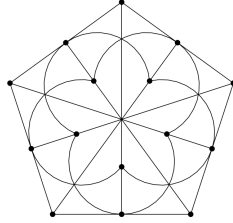


FIGURE 9. The Doily

We now consider a construction of a family of generalized quadrangles which we will use later.

Example 4.53. Let $q = 2^h$. Consider the projective space $PG(3, q)$, a projective plane π embedded in $PG(3, q)$ and a hyperoval $\mathcal{H}$ in π . Then define $T_2(\mathcal{H})$ to be the incidence geometry whose points are all the points in $PG(3, q)$ not contained in π and lines are the lines not contained in π and cross $\mathcal{H}$ through a point. $T_2(\mathcal{H})$ is a generalized quadrangle of order $(q - 1, q + 1)$.

Proof. Clearly $T_2(\mathcal{H})$ is a partial linear space since it is a subgeometry of a linear space. Let $T_2(\mathcal{H}) = (\mathcal{P}, \mathcal{L}, \mathcal{I})$ and consider $p \in \mathcal{P}, l \in \mathcal{L}$ such that $p \notin l$. The span of l and p produces a plane which will intersect π at a line l' . Since $\mathcal{H}$ is a hyperoval l' must intersect $\mathcal{H}$ at exactly two points, one will be $x \in l \cap \mathcal{H}$ and the other is some other unique point $y \in \mathcal{H}$. The line py will be in the span of l and p , which is a projective plane, thus intersecting l at some unique point p' . It is clear that $p' \in l$ will be the unique point collinear to p as otherwise l' would contain three distinct points of the hyperoval. Furthermore, through every point must cross at least 2 lines since there are $q + 2 > 2$ distinct points in $\mathcal{H}$. Any line of $T_2(\mathcal{H})$ is incident to q points as we have removed all the points of the hyper oval $\mathcal{H}$ and all points are incident to $q + 2$ lines, one for each $h \in \mathcal{H}$. $\square$

5. RAMSEY CONSTRUCTIONS

Using the many tools we developed, let us provide some bounds on Ramsey numbers. We begin by a basic result for an upper bound of $R(k, k)$.

Theorem 5.1. *For any $k \in \mathbb{N}$, $R(k, k) \leq 4^k$.*

Proof. Let $n = 4^k$ and consider an arbitrary edge coloring of K_n . We proceed by constructing a sequence of vertices $(v_i)_{i=1}^{2k}$ as follows. Pick $v_1 \in V$ a vertex of K_n arbitrarily, let R_1, B_1 be the set of vertices connect to v_1 via a red or a blue edge respectively. One of the two sets has cardinality at least 2^{2k-1} , say R_1 , pick $v_2 \in R_1$ arbitrarily from that set. Iteratively apply the process using v_2 and the subgraph induced by R_1 to obtain the sequence $(v_i)_{i=1}^{2k}$. Among these $2k$ step we must have picked at least k times the same color say at the vertices $(v_{\phi(i)})_{i=1}^k$, the complete graph induced by these elements is monochromatic. $\square$

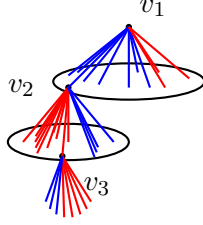


FIGURE 10. Picking vertices

5.1. Probabilistic constructions. The simplest and most powerful technique to quantitative results on Ramsey numbers has for now been the probabilistic method. Let us first derive some lower bounds on $R(k, k)$ using the method.

Theorem 5.2. *For all $k \in \mathbb{N}^*$ the Ramsey number $R(k, k) > \frac{k2^{(k-1)/2}}{e}$.*

Proof 1. Let $n = \frac{k2^{(k-1)/2}}{e}$ and consider a random edge coloring of K_n where an edge is colored independently either blue or red with equal probability. Let us show the probability of containing a monochromatic K_k is less than 1:

$$\begin{aligned}
 \mathbb{P}\left[\bigcup_{\substack{S \subseteq [n] \\ \#S=k}} \{K_S \text{ mono.}\}\right] &\leq \sum_{\substack{S \subseteq [n] \\ \#S=k}} \mathbb{P}[\{K_S \text{ mono.}\}] \\
 &= 2 \binom{n}{k} \left(\frac{1}{2}\right)^{\binom{k}{2}} \\
 &\leq 2 \frac{n^k}{k!} \left(\frac{1}{2}\right)^{\binom{k}{2}} < \left(\frac{en}{k2^{(k-1)/2}}\right)^k = 1
 \end{aligned}$$

Where we used the inequality $k! > 2 \left(\frac{k}{e}\right)^k$ on the last inequality. We deduce there must exist an edge coloring with no monochromatic K_k subgraph. $\square$

Proof 2. We choose the same random coloring as in the first proof and let $n = \frac{k2^{(k-1)/2}}{e}$. Consider $X := \#\{\text{monochromatic } K_k \subset K_n\}$, which is written as $X = \sum_{S \in \binom{[n]}{k}} X_S$, where for a k -subset $S \subset [n]$, define $X_S := \mathbb{1}[K_S \text{ mono.}]$. By linearity we find $\mathbb{E}[X] = \sum_{\substack{S \subset [n] \\ \#S=k}} \mathbb{P}[K_S \text{ mono.}]$ and with the same analysis as above, we deduce $\mathbb{E}[X] < 1$. Thus there exists a coloring such that $X < 1$, as $X \in \mathbb{N}$, we find $X = 0$ and conclude that there is a coloring containing no monochromatic K_k . $\square$

The next theorem uses a method called *alteration*, which in essence considers a random structure with will hopefully has some desired properties and in the case where some properties are missing, we may alter the random structure to add them explicitly.

Theorem 5.3. *For any $n, k \in \mathbb{N}$, $R(k, k) > n - \frac{\binom{n}{k}}{2^{\binom{k}{2}-1}}$.*

Proof. As before consider a random edge coloring of K_n where an edge is colored independently either blue or red with equal probability and let X be defined as above. We have seen that $\mathbb{E}[X] = \binom{n}{k} 2^{1-\binom{k}{2}}$, thus there exists a coloring such that K_n contains at most $\binom{n}{k} 2^{1-\binom{k}{2}}$ monochromatic K_k . Now let $S = \{x_1, x_2, \dots, x_m\}$ be m distinct vertices, one from each of the colored cliques and consider the induced complete subgraph, G' , obtained by removing S . We conclude that the coloring of G' will contain no monochromatic K_k and have at least $n - \binom{n}{k} 2^{1-\binom{k}{2}}$ vertices, proving the statement. $\square$

We can make a small improvement on 5.2. Choosing $n \sim \frac{1}{e} k 2^{k/2} (1 - o(1))$ we find $R(k, k) > \frac{k 2^{k/2}}{e} (1 + o(1))$, a careful analysis shows this simple alteration method doesn't reach any further. The current best lower bound of $R(k, k)$ is proven in [5] and makes use of Lovász local lemma, however the method only makes the improvement $R(k) \geq k 2^{k/2} (\sqrt{2}/e + o(1))$.

5.2. Explicit constructions. Let us try and construct graphs to obtain bounds on $R(3, t)$, first we give an equivalent formulation of the Ramsey numbers $R(s, t)$:

$R(s, t) \in \mathbb{N}$ is the minimal integer such that for any graph G on $m \geq R(s, t)$ vertices, G either contains an independent set of size s or a clique of size t .

The problem of bounding $R(3, t)$ from below asks construct triangle free graphs with independence number less than t . It is known that $R(3, t) = \Theta(\frac{t^2}{\log t})$ as shown in [6], but the proof relies on technical probabilistic arguments, instead we shall construct graphs which get us just a $\log t$ factor away

from the actual estimate. We begin with a simple approach employing the projective plane of order q .

Theorem 5.4. $R(3, t) = \Omega(t^{3/2})$

Proof. We shall consider the incidence graph $G = (\mathcal{P} \cup \mathcal{L}, E)$ of the projective plane $PG(2, q)$ and an arbitrary ordering $>$ of the edges E . Denote $H_>$ the graph whose vertices are the edges of G and two vertices $pl < p'l'$ with $p \neq p', l \neq l'$ are joint by an edge if and only if $pl' \in E$. $H_>$ must be triangle free. Suppose it weren't, then there exists $p_1l_1 < p_2l_1 < p_3l_3$ such that all the nodes are joint by an edge, thus $p_1l_2, p_1l_3, p_2l_3 \in E$, however, since $PG(2, q)$ is a linear space and p_1, p_2 lie on two common lines l_2, l_3 this isn't possible. Furthermore, $H_>$ cannot contain independent set of size more than $m := 2(q^2 + q + 1) - 1$. Suppose there was an independent set $p_0l_0 < \dots < p_ml_m$, as these are $m + 1$ edges in the connected $m + 1$ vertex graph G , it must form a cycle of length $2k \geq 4$. Let $(p_{\phi(1)}, l_{\phi(1)}, \dots, p_{\phi(k)}, l_{\phi(k)})$ be that cycle, there must an index i such that $p_{\phi(i+1)}l_{\phi(i+1)} < p_{\phi(i)}l_{\phi(i)}$, where $i + 1$ is taken modulo k . We deduce as $p_{\phi(i+1)}l_{\phi(i)} \in E$ that $p_{\phi(i+1)}l_{\phi(i+1)}$ and $p_{\phi(i)}l_{\phi(i)}$ are joint by an edge in $H_>$, a contradiction. Thus we have constructed a triangle free graph on $(q+1)(q^2+q+1)$ vertices with no independent set of size $2(q^2+q+1)$ implying $R(s, q) > Cq^{3/2}$ for some constant C . Of course we may now generalize the result for any integer t by bounding it as $2(q^2 + q + 1) \leq t < 4(q^2 + q + 1)$ for some prime power q . $\square$

Now for a second more technical construction we will employ generalized quadrangles, which, by nature, don't have geometric triangles. We may translate this property to graphs by considering the associated collinearity graph of the geometry, which will only contain triangles whenever three points lie on a common line. Therefore, we may hope to construct a generalized quadrangle which is triangle free and whose collinearity graph has a low independence number. For this we introduce Kopparty's graph.

We consider the generalized quadrangle $T_2(\mathcal{H})$ of order $q = 2^h$, as described in 4.53, with the hyperoval $\mathcal{H} = \{(0, 1, u, u^2) : u \in \mathbb{F}_q\} \cup \{(0, 0, 0, 1), (0, 0, 1, 0)\}$ embedded in the plane $\pi = \{(0, 1, x_3, x_4) \in PG(3, q)\}$. Now the points of the geometry are given by $(1, x_2, x_3, x_4) \in PG(3, q)$ and the lines are all lines crossing through $\mathcal{H}$ not contained in π . Consider the subgeometry induced by removing the lines crossing through $(0, 1, 0, 0), (0, 0, 1, 0)$ or $(0, 0, 0, 1)$ and we let G be the collinearity graph associated to the geometry. Adjacency of two points may be checked algebraically as $(1, x_2, x_3, x_4) \sim (1, y_2, y_3, y_4)$ if and only if $\exists a, u \in \mathbb{F}_q^*, (0, x_2 - y_2, x_3 - y_3, x_4 - y_4) = (0, a, au, au^2)$. Finally we consider the subgraph G' of G where two nodes are adjacent with the extra condition $\text{tr}(au^{-1}) = 1$, where the trace function is taken to be $\text{tr} : \mathbb{F}_q \rightarrow \mathbb{F}_2$ with $\text{tr}(x) = \sum_{i=1}^{q-1} x^i$. G' is called Kopparty's graph.

Proof 2. Consider Kopparty's graph G' . First let us show G' is triangle free.

By the discussion above, as G' is a subgraph induced by the subgeometry of a generalized quadrangle, we know the only possible triangles in G' arise from three vertices x, y, z all on a common line, suppose that such an x, y, z forms a triangle. Then, $x - y = (0, a, au, au^2)$, $z - x = (0, b, bu, bu^2)$ and $y - z = (0, c, cu, cu^2)$ for some $a, b, c, u \in \mathbb{F}_q^*$, however by linearity of the trace, we find $\text{tr}((a + b + c)u^{-1}) = 1$ which is not possible as $(a + b + c) \cdot (0, 1, u, u^2) = (x - y) + (y - z) + (z - x) = 0$. We conclude that G' must be triangle free.

Now let us show that $\alpha(G')$ is not too large. From the algebraic condition for two vertices being joined we notice that G' is isomorphic to the Cayley graph of the additive group $\mathbb{F}_q^3 \cong \mathbb{F}_2^{3h}$ with the generating set $S := \{((a, au, au^2) : a, u \in \mathbb{F}_q^* \text{ tr}(au^{-1}) = 1\}$. Now we note by 3.7, that the eigenvalues of G' are $\lambda_u = \sum_{s \in S} (-1)^{\text{tr}(s \cdot u)}$ where u runs over $\mathbb{F}_q^3$. We claim that G' is a $(q^3, (q-1)^2, 3q)$ -graph. It is clear that the number of vertices of G' is q^3 and as a Cayley graph will have regularity constant $\#S = (q-1)^2$, we now prove that the eigenvalues other than the eigenvalue of regularity $d = (q-1)^2$ are bounded by $3q$. Let $u = (u_1, u_2, u_3) \in \mathbb{F}_q^3$ and $\lambda = \sum_{s \in S} (-1)^{\text{tr}(s \cdot u)}$ be an eigenvalue different from d , which implies that u is non-zero. We may write $\forall s \in S, s = (a, ax, ax^2)$ the dot product $u \cdot s = a(u_1 + u_2x + u_3x^2)$, we denote $f(x) = u_1 + u_2x + u_3x^2$. We rewrite the eigenvalue as:

$$\lambda_u = \sum_{\substack{a, x \in \mathbb{F}_q^* \\ a \cdot x^{-1} \in \text{tr}^{-1}(1)}} (-1)^{\text{tr}(af(x))} = \sum_{x \in \mathbb{F}_q^*} \sum_{y \in \text{tr}^{-1}(1)} (-1)^{\text{tr}(yxf(x))}$$

We shall show that for a fixed $x \in \mathbb{F}_q^*$ if $xf(x) =: g(x) \notin \mathbb{F}_2$, the inner sum must cancel to 0. Well if $g(x) \notin \mathbb{F}_2$ then the linear forms $z \mapsto \text{tr}(z)$ and $z \mapsto \text{tr}(g(x)z)$ will have distinct kernels let $t \in \mathbb{F}_q^*$ be in the kernel of the first form and not such that $\text{tr}(tg(x)) = 1$, then we observe that $(-1)^{\text{tr}(tg(x))} \sum_{y \in \text{tr}^{-1}(1)} (-1)^{\text{tr}(yxf(x))} = \sum_{y \in \text{tr}^{-1}(1)} (-1)^{\text{tr}(yxf(x))}$ allowing us to conclude the claim. Thus, we conclude that $\lambda_u = \sum_{x \in g^{-1}(\mathbb{F}_2)} \sum_{y \in \text{tr}^{-1}(1)} (-1)^{\text{tr}(yg(x))}$ but note that the number of solutions to $g(x) = c$ for any constant $c \in \mathbb{F}_q$ is at most three as $g(x)$ is a non constant cubic polynomial. We may conclude by the triangle inequality that:

$$|\lambda_u| = \left| \sum_{x \in g^{-1}(\mathbb{F}_2)} \sum_{y \in \text{tr}^{-1}(1)} (-1)^{\text{tr}(yg(x))} \right| \leq 6\#\text{tr}^{-1}(1) = 3q$$

Thus as a $(q^3, (q-1)^2, 3q)$ -graph, using 3.14, we deduce $\alpha(G') \leq \frac{3q^4}{(q-1)^2}$ and thus concluding that $R(3, t) = \Omega(t^{3/2})$. $\square$

6. CONCLUSION

We have explored many areas of combinatorics and how they can all come together in the study of Ramsey numbers. Leveraging, probabilistic, geometric and algebraic arguments we have constructed graphs interesting in bounding Ramsey numbers. Some aspects to look deeper into are the many other constructions of Ramsey Graphs and the so called *Book Algorithm* which has recently obtained an exponential improvement on the upper bound of $R(k, k)$ presented in 5.1. I am pleased with the many things I learned and the mathematical culture I have acquired throughout the project.

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